City of Givers and Takers - Iterated Prisoner's Dilemma

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1 Background and Research Aims

Communities of similar size and resources can end up in very different places: some produce a lot of wealth and share it widely, while others produce less and concentrate it. A common explanation is social capital, people's willingness to cooperate rather than exploit, yet across 957 estimates from 83 studies its average effect on economic growth is close to zero, with large positive and negative effects in different places [1]. Who is connected to whom appears to matter however: in data on 21 billion Facebook friendships, the share of high-income friends that low-income people have is one of the strongest predictors of upward mobility, while how tight-knit a network is shows little association with it [2].

Whether cooperation makes a community more equal is unclear. Network-game models find inequality falling as cooperation rises [3, 4], laboratory experiments find that clustered networks produce more inequality than random ones and separate rich from poor players [5], and making wealth visible increases early differences [6]. No real community can be held fixed while who meets whom is varied, but a model can, and because these outcomes come from many simple local interactions, an agent-based model is the natural tool [7]. We ask: in simulated cities that differ in their mix of cooperative citizens (*givers*) and exploitative citizens (*takers*), what patterns emerge in prosperity, in how that prosperity is distributed, and in who ends up connected to whom? We test four hypotheses, each with a decision rule that can support or refute it (Appendix C).

1. **H1, prosperity threshold:** wealth per person rises non-linearly with the share of givers, with a share below which cooperation cannot get established.
2. **H2, prosperity and equality decouple:** unlike in earlier network models [3, 4], inequality flattens or rises in the richest cities, because cooperation relies on clusters [8, 9] and visible success attracts more of the surplus [6].
3. **H3, emergent segregation:** with no rule telling anyone to avoid anyone, givers tie to givers and takers are left with fewer, weaker ties.
4. **H4, the poverty trap is equal:** taker-dominated cities are poor but not unequal, so median wealth separates good cities from bad better than inequality does.

2 Originality and Contribution

The Prisoner's Dilemma is a classic problem in game theory whose single-round Nash equilibrium is mutual defection, even though two cooperators both do better than two defectors [10]. When the same two players meet repeatedly and remember previous games, reciprocal strategies become possible and cooperation viable [11], as Axelrod's tournaments showed [12]. The iterated game is also a taught example: Downey's simulation pairs agents by random shuffling, lets strategies evolve and reports mean fitness [7, Ch. 12]. Sugarscape, another, produces unequal wealth that Downey calls a weak example of emergence, because it can be predicted from the agents' different vision, metabolism and lifespan [7, p. 156].

Our model differs from both in four ways. First, a social network decides who plays whom, and ties strengthen, fade, break and are replaced. Second, strategies are fixed, so the mix of givers and takers is a controlled variable: we ask what a mix produces, not which strategy wins. Third, we measure how wealth is distributed, between and within the two classes; our citizens differ only in strategy and starting ties, so a city in which nobody cheats tests whether inequality can arise from who meets whom alone. Fourth, citizens can stop meeting anyone who has cheated them.

Each mechanism has a precedent, namely tie decay [13], partner rewiring [14, 9], distance-dependent contact [15] and leaving a partner who defects [16], but we have not found them combined, or with the distribution of wealth as the outcome. We add them one at a time, and each step is built to reproduce the one before when its new rule is switched off (checked for S3, S5 and S6), so each change can be traced to a single rule.

3 Model Specification and Justification

The city is a network of N citizens (Figure A1; all figures are in Appendix A), in which a tie means two citizens know each other, not that they meet. The starting network is random (Erdős–Rényi), or scale-free (Barabási–Albert) in one scenario, with mean degree 6; a citizen can hold at most as many ties as it starts with. Each citizen keeps one strategy for the whole run (Table B1). The six giver strategies never defect first; the two takers, Always Defect (AllD) and Random (cooperates half the time), defect unprovoked. A third taker used in one experiment, Joss, plays Tit-for-Tat (TFT) but defects unprovoked with probability 0.1, and is named after an entry in Axelrod's first tournament [17]. The giver fraction f is the share of givers; strategies within a class are split equally and assigned at random.

Each meeting is one game with payoffs $(T, R, P, S) = (5, 3, 1, 0)$ (Table B2), so a pair of cooperators creates 6 units of wealth, a cooperator and a defector 5, and two defectors 2. Payoffs are added to wealth rather than moved from a fixed pool, in which every pairing would create the same total, so total wealth records how citizens treat each other.

Each week every citizen starts one meeting. Citizen i picks partner j with probability proportional to

$$s_{ij} = (w_{ij} + \varepsilon)^{\beta} \, R_j^{\gamma} \, d_{ij}^{-\lambda}, \tag{1}$$

where w_{ij} is their tie strength (0 without a tie), R_j is j's rating, and d_{ij} is the number of steps between them, up to $d_{\max} = 3$. The offset $\varepsilon = 0.1$ lets citizens without a tie meet, which is how new ties form; the $1/d$ shape follows the share of real social ties, which falls roughly as one over geographic distance [15]; and $\beta = \gamma = \lambda = 1$ weight past meetings, reputation and distance equally. Because attention is limited, a citizen can receive at most $c_{\max} = 4$ meetings a week.

Every meeting adds 1 to the tie's strength, and two citizens without a tie form one when they meet, if both have spare capacity. Between meetings a tie fades as $w_{ij} e^{-\alpha \Delta t}$, with $\alpha = 0.05$ per week and Δt the weeks since the pair last met, so ties fade from disuse rather than because of the last game [13]. A tie weaker than $\theta = 0.1$ breaks, which takes 46 weeks without a meeting for a tie of strength 1, and citizens with spare capacity then form new ties, mostly with friends of friends. A citizen's rating is

$$R_j = R_{\min} + (R_{\max} - R_{\min}) \, \kappa_j \, \sigma_j, \tag{2}$$

where κ_j is a moving average of the share of j's recent moves that were cooperative, σ_j is j's total tie strength divided by the largest in the city, $R_{\min} = 0.1$ and $R_{\max} = 1$. A cooperative citizen with few ties and a well-connected defector both rate poorly. No rule tells anyone to avoid takers; they lose partners only because defecting lowers their rating.

In the last step citizens can refuse: once either citizen in a pair has defected on the other, the pair stops meeting and their tie breaks, permanently or for W weeks. The rule never mentions

class, so a taker is excluded only by the citizens it has cheated, one at a time. Every citizen starts with zero wealth, every tie with strength 1 and $\kappa_j = 0.5$. Strategies never change, wealth is never spent and never affects choices, and the city is closed. The values of ε, α, θ and $c_{\max}$ were chosen to keep a city of half givers away from both collapse and saturation of cooperation. Every random step draws from one seeded generator, so each run can be repeated exactly (Table B3).

4 Experimental Design

The six scenarios add one rule at a time (Figure A1): S1 is a well-mixed city; S2 places it on a fixed random network with partners drawn at random from neighbours; S3 adds fading and breaking ties; S4 adds the choice rule (1), ratings and the cap; S5 adds refilling and is the full city, also run on a scale-free network as S5b; and S6 adds refusal. Before any experiment, the model passes 25 validation checks (Appendix D): S3, S5 and S6 each reproduce the step before exactly when their new rule is switched off; the bookkeeping holds, with every unit of wealth coming from a game and no citizen above its tie capacity or meeting cap; values worked out by hand from the payoffs, such as exactly 3 per game in a well-mixed city of givers, come out of the simulation; and extreme settings, from cities of four citizens to a city of only takers under refusal, keep these rules intact.

Every scenario is run at giver fractions $f \in \{0, 0.2, \ldots, 1\}$ with 5 seeds each: 210 cities of $N = 100$ citizens over $T = 500$ weeks. The same seeds are used in every scenario, so differences between neighbouring steps are paired comparisons over the same starting cities. Prosperity is mean and median wealth at week 500, and inequality is the Gini coefficient of wealth, with the Theil index split between and within the two classes [18]. We also record cooperation, class assortativity, each class's partners and payoff per game, and the share of takers with no ties. In S1 to S5 average income and its Gini are close to settled by week 500 (in S5 the Gini of income is 0.249 at week 250 and 0.248 at week 500), while S6 is still changing.

Each hypothesis is tested on S5, and then on S6, with a decision rule fixed before the runs and 95% bootstrap intervals over seeds [19] (Appendix C); H1, for example, needs a two-segment fit of wealth against f to beat a straight line by more than 10 in BIC [20, 21]. Four further experiments use the same seeds: an all-TFT city with the full choice rule, reputation only, past meetings only and random choice; Watts–Strogatz networks, which change clustering at a fixed number of ties; refusal windows from 1 to 208 weeks; and takers split equally between AllD, Random and Joss. To test how robust the results are, a sensitivity test lowers and raises ten settings one at a time (Table B4), reruns S5 and S6 (1,200 cities) and repeats the hypothesis tests. The notebook that runs every experiment, with an animation and controls to run single cities live, is at `https://github.com/Precipicee/cits4403-project` (`notebooks/Project.ipynb`).

5 Results

Prosperity and the steps of the model: More givers make a city richer in every scenario (Figure A2). In S5, wealth per person rises from 1,918 with no givers to 3,000 with only givers, faster once givers are common; a bent line fits better than a straight one (BIC difference 18, bend at $f = 0.29$), so H1 is supported. Across neighbouring steps (Figure A3), a fixed network raises the Gini by 0.119 without changing cooperation, fading ties and refilling change nothing measurable, and partner choice raises wealth per person by 219, the cooperation rate by 0.109 and the Gini by 0.031. Because every citizen prefers well-rated partners, a taker then plays 76% of its games against givers (at $f = 0.5$).

Inequality and connections without refusal: In S5 the Gini rises from 0.136 with no givers to 0.230 at 40% givers, stays level, and falls to 0.210 with only givers, so H2 is inconclusive. Takers end with 2.2 fewer partners than givers, but class assortativity is slightly negative (-0.02

to −0.16 in S4 and S5) and does not rise, so H3 is inconclusive. Taker-dominated cities are poorer (median wealth 573 lower) and more equal (Gini 0.060 lower), so H4 is supported.

Most of this inequality does not come from cheating (Figure A4). A city in which everyone plays TFT, where nobody is cheated, has a Gini of 0.210, against 0.226 for a mixed city. With reputation only it is 0.132, with past meetings only 0.052, and with random choice 0.049. The ranking is set early: games played by week 25 match the final ranking with a rank correlation of 0.80, while starting ties predict it at 0.66. Lowering clustering from 0.600 to 0.054 barely changes cooperation (−0.0075, standard error 0.0066).

Refusal: Permanent refusal (S6) protects givers and cuts takers off (Figures A5 and A6). A giver ends with about 3,000 at every mix (3,057 at $f = 0.2$, against 2,721 without refusal), while by week 500, 97% to 100% of takers in mixed cities have no ties. A taker that nobody meets creates no wealth, so cities with many takers become much poorer (842 per person at $f = 0.2$, against 2,093) and more unequal (Gini 0.564, against 0.184). Takers form no group of their own, because they also defect on each other. The bend of H1 reverses, as each extra 20% of givers adds a little less wealth than the one before, and H4 is refuted: taker-dominated cities are poorer (median wealth 2,091 lower) and more unequal (Gini 0.049 higher). H3 cannot be tested once takers hold no ties. Joss largely escapes refusal, because it cooperates for about ten games with each new partner before it cheats: refusal cuts AllD's wealth from 1,467 to 383 but Joss's only from 1,876 to 1,720 (Figure A7).

How long a refusal lasts: A one-week refusal barely changes the Gini at $f = 0.2$ (0.184 to 0.199), but the share of inequality between givers and takers rises from 0.19 to 0.84 (Figure A8). Refusals longer than 52 weeks cost the city wealth (at $f = 0.5$, 2,409 per person at 52 weeks and 1,724 when permanent), as takers run out of willing partners within reach. Similar Gini values hide different cities (Figure A9): the share of inequality between the classes is 0.004 to 0.099 in S2, 0.19 to 0.38 in S5 and 0.57 to 0.94 in S6.

Sensitivity to the settings: Across the 20 changed settings (Figure A10, Table C1), refusal always cuts off 97% to 100% of takers and makes cities with 20% givers 48% to 65% poorer, wealth always rises with the giver share, and a city of givers is always unequal (Gini 0.080 to 0.313, against 0.013 in S1). The bend of H1 always goes upward, but passes the test in 15 settings and moves between $f = 0.07$ and 0.50. H4 passes in 11 and is refuted in 3, three of the four settings where a city of givers is about half as unequal as the most unequal mixed city.

6 Discussion

Prosperity: More givers always make a city richer, which follows from the payoffs: a giver creates the most wealth with another giver, and giver pairs grow faster than givers. This supports H1 with our settings and fits the observation that the benefit of cooperation depends on context [1], but where the rise speeds up depends on the settings, so it is not a real-world threshold, and no city collapses into defection below it, because citizens never change strategy.

How prosperity is shared: Without refusal, most inequality comes from how often citizens are met, not from cheating: being met often raises a citizen's rating, which gets it met more. People show a similar process: when participants in an online music market could see what others had chosen, success became more unequal and less predictable [22]. The size of the effect here comes from our rating rule, as the sensitivity test shows. H2 is inconclusive because inequality comes from attention, which the giver mix changes little, rather than from clusters, which matter little when nobody copies a successful neighbour. Whether poor cities are also

equal (H4) depends on how strongly reputation decides who meets whom. A short refusal moves almost all inequality between the classes, so that the poorest citizens become the takers, while the Gini barely moves; so a Gini alone does not show who holds the inequality.

Who ends up connected to whom: Without refusal, takers are not pushed to the edge, because everyone seeks well-rated partners and givers cannot decline, whereas real people cut ties with those who defect [14]. With refusal, takers are excluded as a class but form no group of their own, unlike real exploiters, who often cooperate with each other. In experiments, excluding selfish members reduced exploitation and the excluded went on to cooperate, and people used gossip to avoid partners who had been selfish with others [23]. Our takers cannot change and wealth comes only from meeting, so exclusion here is more permanent and costly than in a real city.

Emergent behaviour: Downey defines an emergent property as "a characteristic of a system that results from the interaction of its components, not from their properties" [7, pp. 155–156]. By that definition, inequality in a city where nobody cheats, the rise in cooperation under partner choice and refusal although no citizen changes strategy, and the exclusion of takers as a class by a rule that never mentions class are emergent. The rise in prosperity is not, because it follows from the payoffs, and cooperation does not evolve, because strategies are fixed. Unlike Sugarscape's, the inequality in our all-TFT city cannot be predicted from differences between citizens, since starting ties predict the final ranking only partly.

7 Conclusions

We asked what patterns emerge in prosperity, in how it is distributed, and in who ends up connected to whom, as the mix of givers and takers changes. Prosperity follows the mix: more givers always make a city richer, faster once givers are common (H1, supported with our settings). How wealth is shared depends less on the mix than on who is met: most inequality comes from differences in attention rather than from cheating, so inequality levels off instead of falling steadily (H2, inconclusive), and taker-dominated cities are also more equal (H4, supported with our settings), though this depends on how strongly reputation decides who meets whom. Who ends up connected depends on refusal: without it, givers and takers stay mixed (H3, inconclusive); with it, takers are excluded as a class, and cities with many takers become poorer and more unequal, which refutes H4. The model passes all 25 validation checks, and the effects of refusal and the inequality without cheating hold in every setting we changed, while H1 and H4 depend on the settings.

8 Limitations and Future Work

Limitations: Five seeds per setting resolve the large effects of partner choice and refusal, but not small ones such as the effect of clustering on cooperation. The verdicts are the same at 50 and 200 citizens except H4 on S6 at 200, but S6 is still changing at week 500. Refusal is perfect and personal: every defection is noticed, none is a mistake, and nobody hears about a cheat from anyone else. Wealth is never spent and never affects choices, and the cities are closed.

Future work: Letting citizens copy successful strategies would test whether refusal helps cooperation spread, as walking away does in evolutionary simulations [16]. Refusal by reputation could stop a taker like Joss before it cheats each new partner. Mistakes in refusal, a cost of living and comparison with measured social networks [2] would bring the model closer to real cities.

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A Figures

Unless a caption says otherwise, each point is the mean of 5 seeds and error bars are one standard error of the mean.

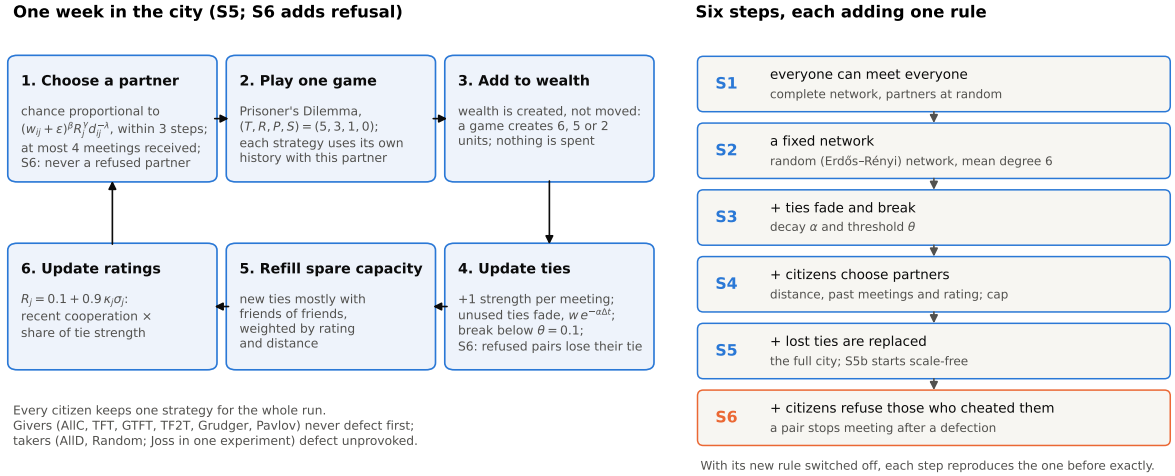


Figure A1: The model. Left: the six updates applied every week, in order; all citizens decide at the same time from the state at the start of the week. Right: the six scenarios, each adding one rule to the one above. S5b is S5 on a scale-free starting network.

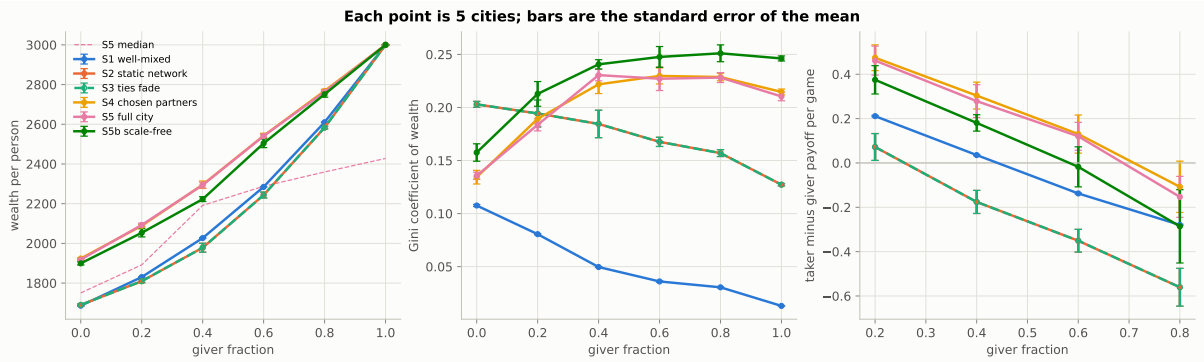


Figure A2: Outcomes at week 500 against the giver fraction, one line per scenario without refusal; each point is the mean of 5 cities and bars are one standard error. Left: wealth per person in payoff units, with the median of S5 dashed. Centre: Gini coefficient of wealth (0 is perfect equality). Right: a taker's mean payoff per game minus a giver's, defined only where both classes are present. S2 and S3 produce identical cities; S3 is drawn dashed over S2.

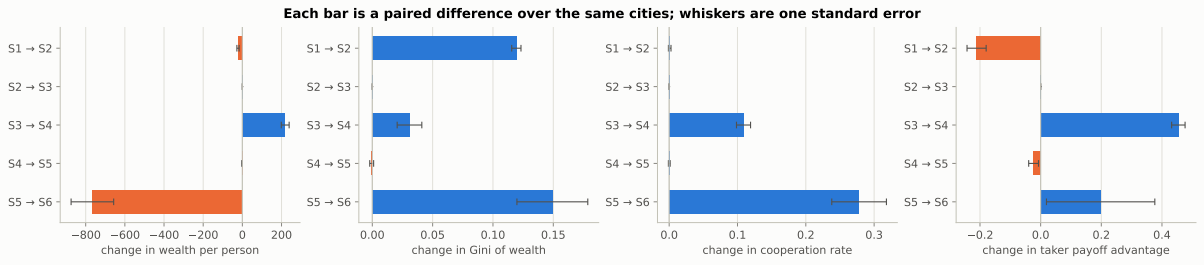


Figure A3: What each step of the model changes. Each row is one step and each bar the mean change in a measure when the same 30 cities (6 giver fractions, 5 seeds) are run with and without that step's rule; blue is an increase and orange a decrease. Left to right: wealth per person, Gini of wealth, cooperation rate (share of cooperative moves) and a taker's payoff per game minus a giver's.

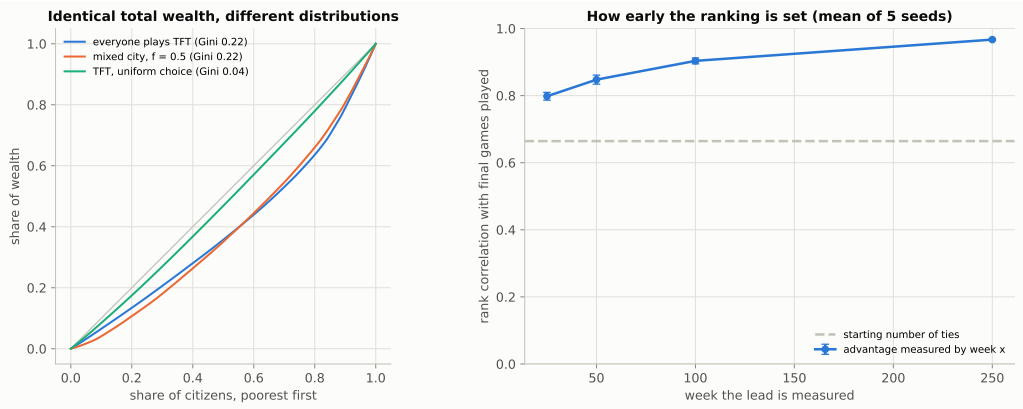


Figure A4: Inequality when nobody is cheated. Left: Lorenz curves for three single cities, the share of all wealth (vertical) held by the poorest share of citizens (horizontal); the diagonal is perfect equality. A city where everyone plays TFT is about as unequal as a mixed city ($f = 0.5$); with partners drawn at random it is close to equal. Right: in the all-TFT city, the rank correlation between games played by a given week and by week 500 (1 means the order is already fixed), mean of 5 seeds with one standard error; dashed, the correlation with the starting number of ties.

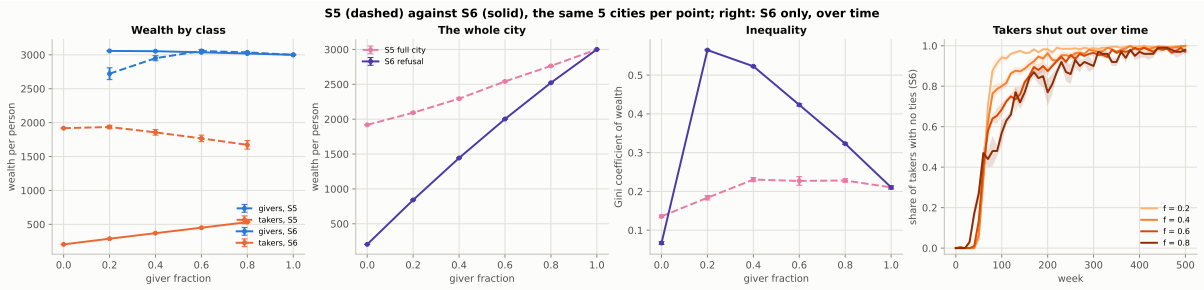


Figure A5: The same cities without refusal (S5, dashed) and with permanent refusal (S6, solid). First three panels, at week 500 against the giver fraction: wealth per giver (blue) and per taker (orange); wealth per person; Gini coefficient of wealth. Right: S6 only, the share of takers with no ties against the week, one line per giver fraction, with a band of one standard error. Means of 5 cities per point.

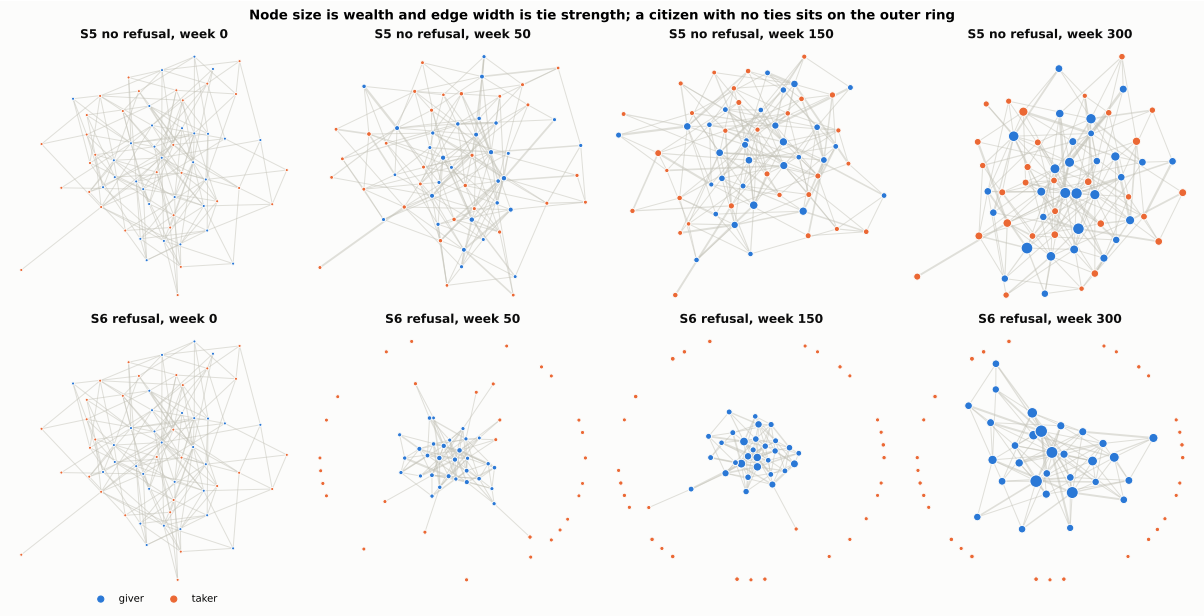


Figure A6: One city of 60 citizens with half givers at weeks 0, 50, 150 and 300, without refusal (S5, top) and with refusal (S6, bottom), from the same starting network and seed. Blue citizens are givers and orange are takers; node size is wealth and line width is tie strength, on the same scales in every panel. A citizen with no ties sits on the outer ring. The animation of this city is in the notebook.

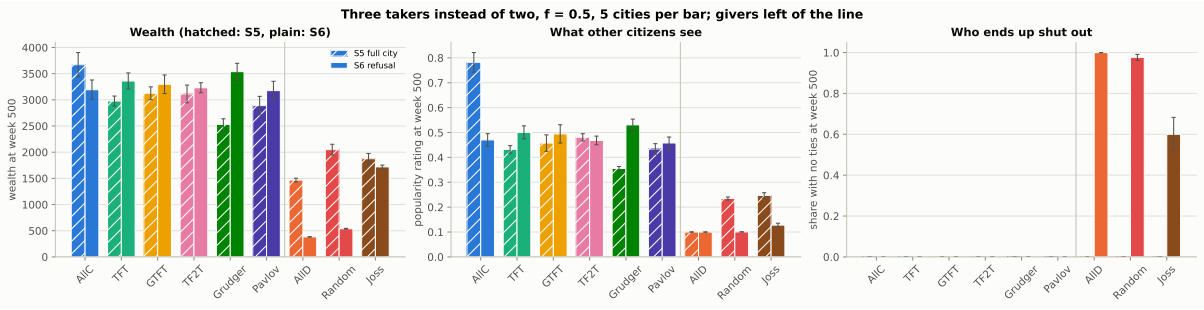


Figure A7: A taker that cooperates most of the time. One pair of bars per strategy at $f = 0.5$, with the takers split equally between AllD, Random and Joss; givers are left of the vertical line, hatched bars are S5 and plain bars S6. Left to right, at week 500: wealth; popularity rating, the only information other citizens have; share holding no ties.

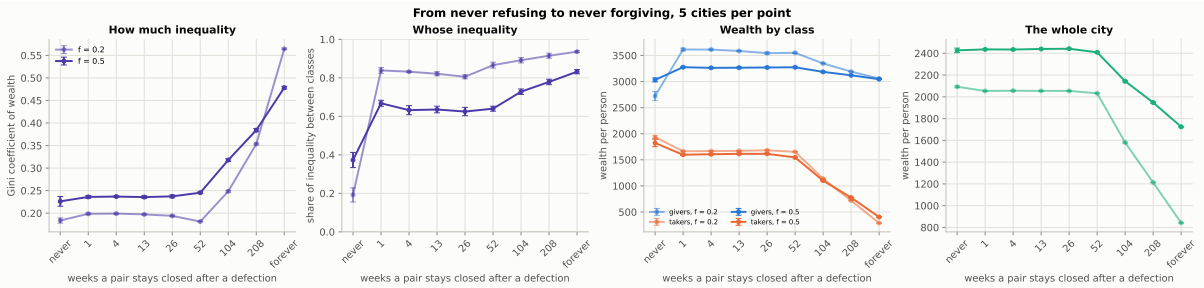


Figure A8: How long a refusal lasts. The horizontal axis is the number of weeks a pair stays closed after a defection, from never (S5) to permanent (S6), not evenly spaced; light lines have 20% givers and dark lines 50%. Left to right, at week 500: Gini of wealth; share of inequality between givers and takers (Theil index, 0 none, 1 all); wealth per giver (blue) and per taker (orange); wealth per person.

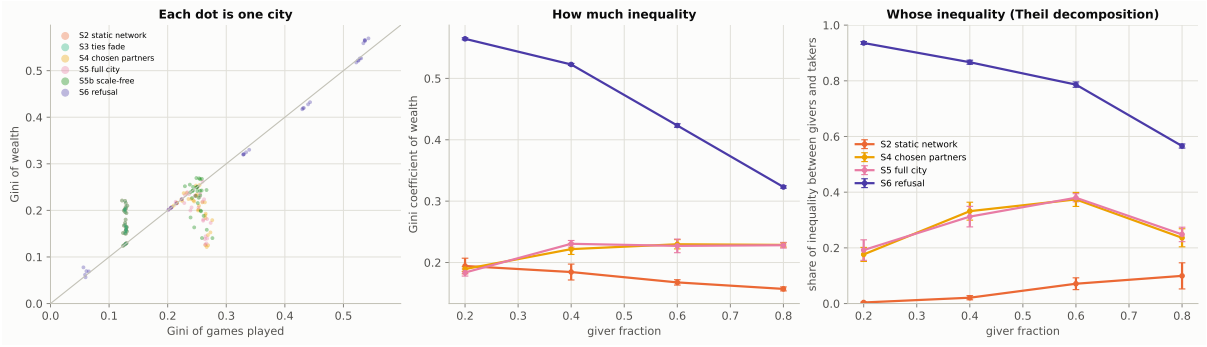


Figure A9: What the Gini coefficient reflects. Left: one dot per city, the Gini of games played (horizontal) against the Gini of wealth (vertical); on the diagonal, wealth is exactly as unequal as the number of games played. Centre: Gini of wealth against the giver fraction. Right: share of inequality between givers and takers (Theil index), for S2, S4, S5 and S6.

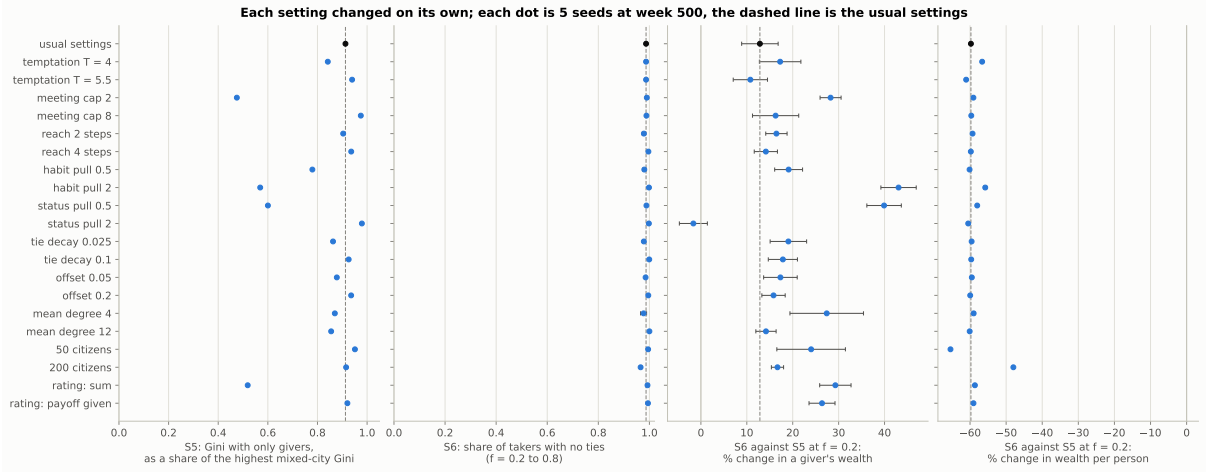


Figure A10: Sensitivity of four results to the settings, one row per changed setting, at week 500; the usual settings are at the top in black and the dashed line marks their value. Each dot is the mean of 5 seeds, with one standard error where it can be calculated seed by seed. Left to right: in S5, the Gini of a city of givers divided by the highest Gini of a mixed city ($f = 0.2$ to 0.8); in S6, the share of takers with no ties ($f = 0.2$ to 0.8); and the change from S5 to S6 at $f = 0.2$, paired by seed, in a giver's wealth and in wealth per person (%).

B Model Reference

Table B1: Strategies. Each citizen is given one at the start and keeps it for the whole run.

Strategy	Behaviour	Class
AllC	Always cooperates	Giver
TFT	Cooperates first, then copies the partner's last move	Giver
GTFT	As TFT, but forgives a defection with probability 0.1	Giver
TF2T	Defects only after two defections in a row	Giver
Grudger	Cooperates until the partner first defects, then always defects	Giver
Pavlov	Repeats its last move if the partner cooperated, otherwise switches	Giver
AllD	Always defects	Taker
Random	Cooperates with probability 0.5	Taker
Joss	As TFT, but defects unprovoked with probability 0.1 (Joss runs only)	Taker

Table B2: Prisoner's Dilemma payoffs to the row player, in units of wealth per game.

	Partner cooperates	Partner defects
You cooperate	$R = 3$	$S = 0$
You defect	$T = 5$	$P = 1$

Table B3: Model parameters and the values used in every reported run, with the configuration key that sets each.

Symbol	Meaning	Value	Configuration key
N	citizens per city	100	`population.n_agents`
T	weeks per run	500	`run.n_rounds`
f	giver fraction	0, 0.2, …, 1	`population.giver_fraction`
k	mean degree of the starting network	6	`topology.mean_degree`
(T, R, P, S)	payoffs	(5, 3, 1, 0)	`game.payoff`
–	GTFT forgiveness probability	0.1	`strategies.gtft_forgiveness`
–	Random cooperation probability	0.5	`strategies.random_cooperation`
–	Joss unprovoked defection probability	0.1	`strategies.joss_defection`
B	meetings started per citizen per week	1	`encounter.initiated`
$c_{\max}$	cap on meetings received per week	4	`encounter.max_received`
$d_{\max}$	furthest candidate partner (steps)	3	`encounter.d_max`
β, γ, λ	pull of past meetings, reputation, distance	1, 1, 1	`encounter.*_exponent`
ε	offset allowing contact without a tie	0.1	`encounter.eps`
–	refusal rule (S6)	experience	`encounter.refusal`
W	weeks a pair stays closed after a defection (S6)	permanent	`encounter.refusal_memory`
w_0	starting tie strength	1	`ties.initial_weight`
δ	strength added per meeting	1	`ties.reinforcement`
α	tie decay rate (per week)	0.05	`ties.decay_rate`
θ	tie breaking threshold	0.1	`ties.severance_threshold`
$R_{\min}, R_{\max}$	rating bounds	0.1, 1	`popularity.r_min`, `r_max`
η	smoothing of the cooperation average	0.1	`popularity.smoothing`
κ_0	starting cooperation average	0.5	`popularity.initial_cooperation`
–	seeds per setting	5	–

Table B4: Settings changed in the sensitivity test, one at a time. Each is halved and doubled where the model allows it. The temptation T must stay between $R = 3$ and $2R - S = 6$, the reach is a whole number of steps, and a random network with mean degree 3 is rarely connected, so these use the nearest values that work.

Setting	Usual value	Lower	Higher
Temptation payoff T	5	4	5.5
Cap on meetings received $c_{\max}$	4	2	8
Reach $d_{\max}$ (steps)	3	2	4
Pull of past meetings β	1	0.5	2
Pull of reputation γ	1	0.5	2
Tie decay α	0.05	0.025	0.1
Offset ε	0.1	0.05	0.2
Mean degree k	6	4	12
City size N	100	50	200
Rating rule	cooperation × ties	cooperation + ties	payoff given × ties

C Hypothesis Decision Rules and Verdicts

Each rule was fixed before the runs. "Clearly" means a bootstrap interval over seeds that excludes zero. Taker-dominated cities have $f \leq 0.3$ and giver-dominated cities $f \geq 0.7$.

1. **H1:** a continuous two-segment fit of wealth against f beats a straight line by more than 10 in BIC, and the slope above the break is steeper, with an interval for the difference above zero. Refuted if the two-segment fit is better by 2 or less, so that wealth is close to a straight line in f.

2. **H2:** somewhere above $f = 0.5$, the Gini and wealth per person both clearly rise between neighbouring giver fractions. Refuted if no step clearly raises the Gini and the rank correlation between f and the Gini is clearly negative.

3. **H3:** class assortativity rises over the last 100 weeks while takers hold fewer partners and weaker ties than givers, in mixed cities. Refuted if assortativity does not rise and takers' partners and tie strength do not differ from givers'.

4. **H4:** taker-dominated cities have lower median wealth than giver-dominated ones and no higher Gini, and the giver fraction explains more of the variation between cities in median wealth than in the Gini. Refuted if their median wealth is not clearly lower or their Gini is clearly higher.

Table C1: Hypothesis verdicts with the usual settings, and in how many of the 20 changed settings of the sensitivity test the verdict is the same.

Hypothesis	S5, no refusal	Same in	S6, permanent refusal	Same in
H1 prosperity threshold	Supported	15 of 20	Inconclusive (bend reverses)	20 of 20
H2 prosperity and equality decouple	Inconclusive	18 of 20	Inconclusive	20 of 20
H3 emergent segregation	Inconclusive	19 of 20	Not testable	n/a
H4 the poverty trap is equal	Supported	11 of 20	Refuted	17 of 20

D Validation Checks

Each check runs in the notebook before any result, mostly on cities of 60 citizens over 120 weeks, and all 25 pass.

1. S3 with no decay and no breaking reproduces S2 exactly, citizen by citizen, including the state of the random number generator.

2. S5 with no breaking reproduces S4 exactly.

3. S6 with refusal switched off reproduces S5 exactly.

4. Every unit of wealth comes from a game payoff.

5. No citizen holds more ties than its capacity, ties are mutual, and nobody is tied to themselves.

6. No citizen receives more meetings than the cap allows.

7. Under refusal, no pair plays again after a defection between them, and no tie survives between two citizens who have refused each other.

8. Joss cooperates with unconditional cooperators 0.906 of the time, against an expected 0.9.

9. Values worked out from the payoffs: a well-mixed city of givers holds exactly 3 per game at 2 games a week; a city of AllD exactly 1 per game; AllC with every move flipped by noise exactly 1 per game; every game creates 2, 5 or 6 units of wealth; and a well-mixed city of AllD and Random earns 1.688 per game, matched within sampling error over 5 cities.

10. Extreme settings: with no takers, S6 reproduces S5 exactly; with only takers and refusal, every citizen ends with no ties and wealth stops growing; cities of 4 and 5 citizens, a cap of one meeting, a one-week refusal, refusal by reputation, near-instant tie decay, almost no contact without a tie, and a scale-free city with refusal all run with the bookkeeping intact; and the same seed reproduces a run exactly.